# PMUN

## BACKGROUND GUIDE - 1

**Topic: Harnessing Solar Energy for Equitable Access and Clean Air**

**Committee: United Nations Environment Programme (UNEP)**

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## A. Understanding the Issue

The world faces a major energy and environmental challenge. Countries across Africa, Asia, Latin America, and parts of Europe continue to rely heavily on fossil fuels like coal, diesel, and biomass for electricity generation and household energy needs. This dependence has contributed to severe air pollution, greenhouse gas emissions, and major public health concerns.

At the same time, millions of people in rural and low-income communities still lack reliable access to electricity. Solar energy has emerged as a promising solution because large parts of the world receive abundant sunlight year-round. Solar power can reduce dependence on fossil fuels, improve air quality, and increase access to clean and affordable electricity.

However, equitable access remains a challenge. Wealthier urban populations and businesses are increasingly adopting solar technology, while poorer communities often struggle due to high installation costs, weak infrastructure, lack of financing, and limited awareness.

## B. Why is this Important Globally?

Clean energy is directly connected to climate change, economic development, public health, and global sustainability. Many regions of the world suffer from dangerous levels of air pollution, and improving access to renewable energy is essential to reducing pollution and achieving international climate goals.

**Public Health:** Air pollution causes millions of premature deaths annually.

**Climate Change:** Fossil fuels are a major source of greenhouse gas emissions.

**Economic Development:** Renewable energy can create green jobs and improve energy security.

**Energy Equality:** Solar energy can bring electricity to remote and underserved communities.

**SDG 7:** The UN aims to ensure affordable and clean energy for all by 2030.

## C. Key Statistics & Trends

- Over 700 million people worldwide still lack access to electricity, mostly in Sub-Saharan Africa and parts of Asia.
- Air pollution contributes to approximately 7 million premature deaths globally every year, according to the WHO.
- Global solar capacity crossed 1,000 GW in 2022 and continues to grow rapidly.
- Countries like Morocco, Kenya, and Chile have become global leaders in solar energy adoption among developing nations.
- The World Bank estimates that transitioning to clean energy could add $26 trillion to the global economy by 2030.

These statistics highlight the strong connection between energy choices, economic growth, public health, and environmental sustainability.

# D. Learn About Your Committee – UNEP

### What Does UNEP Do?

- Coordinates global environmental efforts within the UN system
- Promotes renewable energy and sustainable development
- Provides research, reports, and environmental assessments
- Supports countries in developing environmental policies

### Powers:

- Can recommend environmental policies and frameworks
- Can coordinate international environmental initiatives
- Can encourage global investment in clean energy

### Limitations:

- Cannot directly enforce laws in sovereign nations
- Depends on member states for implementation
- Relies heavily on international cooperation and funding

### Past Actions:

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# E. UN Actions and Frameworks

**I. Stockholm Conference (1972):**

Led to the creation of UNEP and encouraged countries to consider environmental protection alongside development. The Stockholm Conference was the first major international gathering focused entirely on environmental issues. It recognised that economic development and environmental protection must work together rather than against each other.
This conference led to the creation of the United Nations Environment Programme (UNEP), which became the UN's primary environmental body.

### Key Features:

1. First global environmental conference
2. Introduced environmental protection into international politics
3. Led to the creation of UNEP
4. Encouraged countries to begin developing environmental policies

### Why it Matters:

1. Marked the beginning of global environmental cooperation
2. Created international awareness about pollution and sustainability

### Limitations:

1. Focused more on awareness than enforcement
2. Many developing countries were still prioritising industrial growth over environmental protection

**Insight:** Delegates can discuss whether early environmental agreements were strong enough to prevent today's pollution crisis.

## II. Agenda 21 and Earth Summit (1992):

Encouraged countries to adopt sustainable development strategies and cleaner energy systems. The Earth Summit held in Rio de Janeiro focused on sustainable development and environmental responsibility. One of its major outcomes was Agenda 21, a global action plan encouraging countries to adopt environmentally sustainable policies.

The summit highlighted the importance of renewable energy and reducing pollution while balancing economic growth.

### Key Features:

1. Promoted sustainable development globally
2. Encouraged cleaner energy systems
3. Focused on balancing development with environmental protection
4. Increased cooperation between developed and developing countries

### Why it Matters:

1. Helped countries begin integrating renewable energy into development planning
2. Increased international discussion on climate and pollution issues

### Limitations:

1. Agenda 21 was not legally binding
2. Implementation depended entirely on national governments

**Insight:** Delegates may debate whether voluntary environmental cooperation is effective enough.

## III. Kyoto Protocol (1997):

An international agreement aimed at reducing greenhouse gas emissions. The Kyoto Protocol was one of the first international agreements aimed at reducing greenhouse gas emissions. Developed countries were given legally binding emission reduction targets.

Although many South Asian countries were not required to reduce emissions under the agreement, the protocol encouraged cleaner energy adoption and international climate discussions.

### Key Features:

1. Focused on reducing greenhouse gas emissions
2. Introduced legally binding targets for developed countries
3. Encouraged renewable energy investments
4. Promoted international climate responsibility

### Why it Matters:

1. Increased global pressure to move away from fossil fuels
2. Encouraged countries to begin renewable energy planning

### Limitations:

1. Developing countries had fewer obligations
2. Some major countries did not fully participate
3. Emission reductions remained limited globally

**Insight:** Students can debate whether climate responsibility should be shared equally between developed and developing countries.

## IV. Paris Agreement (2015):

Countries are committed to reducing emissions and limiting global warming. The Paris Agreement is one of the most significant international climate agreements. Countries are committed to limiting global warming and reducing greenhouse gas emissions through national climate action plans called Nationally Determined Contributions (NDCs).
South Asian countries included goals for renewable energy expansion and pollution reduction in their climate strategies.

### Key Features:

1. Global agreement to limit temperature rise
2. Countries submit Nationally Determined Contributions (NDCs)
3. Encourages renewable energy expansion
4. Supports international climate cooperation

### Why it matters:

1. Strengthened global climate commitments
2. Increased investment in renewable energy technologies
3. Encouraged countries to set long-term clean energy goals

### Limitations:

1. Countries decide their own targets
2. Enforcement mechanisms are weak
3. Progress depends heavily on political will and financing

**Insight:** Delegates can discuss whether countries are doing enough to meet climate commitments.

## V. Sustainable Development Goal 7:

Focuses on ensuring access to affordable, reliable, sustainable, and modern energy for all. SDG 7 is part of the United Nations Sustainable Development Goals adopted in 2015. It focuses on ensuring access to affordable, reliable, sustainable, and modern energy for all by 2030.
For South Asia, SDG 7 is especially important because millions still lack reliable access to electricity.

### Key Targets:

1. Increase renewable energy use
2. Improve energy efficiency
3. Expand clean energy access
4. Support sustainable infrastructure development

### Why it Matters:

1. Connects energy access with development and poverty reduction
2. Encourages investment in solar and renewable energy
3. Supports long-term environmental sustainability

### Challenges:

1. High infrastructure costs
2. Unequal access between urban and rural populations
3. Slow implementation in some countries

**Insight:** Delegates may debate how developing countries can achieve SDG 7 while managing economic challenges.

### VI. International Solar Alliance (ISA)

The International Solar Alliance was launched by India and France in 2015 to promote the adoption of solar energy, especially in countries with high solar potential. The alliance works to improve financing, technology sharing, and cooperation between countries.

**Key Features:**

1. Promotes global solar energy cooperation
2. Supports research and technology sharing
3. Encourages investment in solar infrastructure
4. Focuses heavily on developing countries

**Why it matters:**

1. Helps countries reduce dependence on fossil fuels
2. Encourages regional cooperation in South Asia
3. Makes solar technology more accessible

**Limitations:**

1. Progress depends on funding and political cooperation
2. Many developing countries still face infrastructure barriers

**Insight:** Delegates can explore whether international partnerships alone are sufficient to ensure equitable access to clean energy.

## F. Country Positions

### I. India:

India is a regional leader in solar energy and co-founded the International Solar Alliance. India is one of the largest renewable energy producers in the world and is considered a regional leader in solar energy development. The country faces severe air pollution challenges, especially in cities such as Delhi, Mumbai, and Kolkata, where fossil fuel dependence and rapid urbanisation have significantly affected air quality.

India launched the National Solar Mission in 2010 to increase solar power generation and reduce dependence on coal. It also co-founded the International Solar Alliance (ISA) with France in 2015 to promote global solar cooperation.

**Key Policies and Initiatives:**

1. National Solar Mission
2. PM-KUSUM Scheme for solar-powered agricultural pumps
3. Rooftop Solar Programme
4. One Sun, One World, One Grid initiative

**National Interests:**

1. Reducing fossil fuel dependence
2. Improving energy security
3. Expanding electricity access
4. Becoming a global renewable energy leader

**Challenges:**

1. Heavy dependence on coal for electricity
2. Unequal access between urban and rural areas
3. Land acquisition issues for solar parks
4. Financing and infrastructure gaps

### II. Kenya:

Kenya is a leading example of solar energy adoption in Sub-Saharan Africa. The country has invested heavily in off-grid solar solutions to bring electricity to rural communities, and is recognised globally for its innovative pay-as-you-go solar financing models.

**Key Policies**

1. Kenya Renewable Energy Policy
2. Rural Electrification Programme
3. Off-grid solar subsidies and financing schemes

**National Interests:**

1. Expanding rural electrification
2. Reducing energy poverty
3. Lowering dependence on expensive diesel generators
4. Improving public health through cleaner energy

**Challenges:**

1. Limited grid infrastructure
2. Financial constraints for low-income households
3. Dependence on imported solar technology
4. Vulnerability to climate-related disruptions

### III. Morocco:

Morocco has become one of the world's most ambitious solar energy leaders, home to the Noor Ouarzazate Solar Complex — one of the largest solar power plants on earth. The country aims to generate 52% of its electricity from renewables by 2030.

**Key Policies**

1. National Energy Strategy 2030
2. Noor Solar Programme
3. Net metering and renewable energy investment frameworks

**National Interests:**

1. Reducing fossil fuel import costs
2. Becoming a clean energy exporter
3. Reducing urban air pollution
4. Attracting international green investment

**Challenges:**

1. High upfront infrastructure costs
2. Unequal access between urban and rural areas
3. Dependence on international financing
4. Need for grid modernisation

### iv) Brazil:

Brazil has a strong renewable energy base, primarily through hydropower, and is increasingly expanding into solar energy. The country receives some of the highest levels of solar irradiation in the world, making it well-positioned for large-scale solar adoption.

**Key Policies**

1. National Energy Plan
2. ProGD Rooftop Solar Programme
3. ANEEL renewable energy auctions

**National Interests:**

1. Diversifying beyond hydropower
2. Expanding electricity access in remote Amazon regions
3. Reducing carbon emissions
4. Creating green jobs

**Challenges:**

1. Unequal energy access between urban and rural regions
2. Regulatory complexity
3. Financing barriers for low-income communities
4. Balancing development with deforestation concerns

### V) Germany:

Germany is a global leader in renewable energy policy and has made significant investments in solar power as part of its Energiewende (energy transition) strategy. It serves as a model for how developed nations can shift away from fossil fuels.

**Key Policies**

1. Energiewende (Energy Transition) Policy
2. Renewable Energy Sources Act (EEG)
3. Solar rooftop subsidies and feed-in tariffs

**National Interests:**

1. Achieving carbon neutrality by 2045
2. Reducing dependence on imported fossil fuels
3. Exporting renewable energy technology
4. Leading global climate policy

**Challenges:**

1. High transition costs
2. Balancing energy affordability with sustainability
3. Grid stability during the transition
4. Managing the phase-out of coal-dependent regions

## G. Case Studies

1. **India – PM-KUSUM Scheme:** Supports solar-powered agricultural pumps and decentralised renewable energy systems.
   Source: *https://pmkusum.mnre.gov.in/*
2. **Kenya – M-KOPA Solar:** A globally recognised pay-as-you-go solar model that has brought affordable solar energy to over 3 million homes across East Africa.
   Source: *https://www.worldbank.org/en/topic/energy/brief/off-grid-solar*
3. **Morocco – Noor Solar Complex:** One of the world's largest concentrated solar power plants, demonstrating how developing nations can lead in large-scale clean energy.
   Source: *https://pmkusum.mnre.gov.in/*
2. **Kenya – M-KOPA Solar:** A globally recognised pay-as-you-go solar model that has brought affordable solar energy to over 3 million homes across East Africa.
   Source: *https://www.iea.org/articles/morocco-a-renewable-energy-frontrunner*

## H. List of Agendas

- Should governments subsidise solar energy for low-income households?
- Can countries cooperate despite political tensions?
- Should solar energy projects prioritise rural and underserved communities?
- How can countries balance economic growth with environmental sustainability?

## I. References & Recommended Reading

**UN & International Bodies**
UNEP: *https://www.unep.org/*
UN SDG 7: *https://sdgs.un.org/goals/goal7*
International Solar Alliance: *https://isolaralliance.org/*
International Energy Agency (IEA): *https://www.iea.org/*
IRENA Renewable Energy Statistics: *https://www.irena.org/*
World Bank Energy Portal: *https://www.worldbank.org/en/topic/energy*

**Country-Specific**
India – Ministry of New and Renewable Energy: *https://mnre.gov.in/*
India – PM-KUSUM Scheme: *https://pmkusum.mnre.gov.in/*
Kenya – Rural Electrification Authority: *https://www.rea.go.ke/*
Morocco – Moroccan Agency for Sustainable Energy (MASEN): *https://www.masen.ma/*
Brazil – National Electric Energy Agency (ANEEL): *https://www.aneel.gov.br/*
Germany – Federal Ministry for Economic Affairs and Climate Action: *https://www.bmwk.de/*

**Reports & Data**
WHO Air Quality & Health: *https://www.who.int/health-topics/air-pollution*
World Bank Off-Grid Solar Market: *https://www.worldbank.org/en/topic/energy/brief/off-grid-solar*
IEA World Energy Outlook: *https://www.iea.org/reports/world-energy-outlook-2023*
IRENA Global Renewables Outlook: *https://www.irena.org/publications*